"An Introduction to Prolog"
Part III

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- This presentation may contain material protected by copyright laws.
- In particular, these lessons are based on the material that Peter Schachte made available on the Web years ago, as part of the Logic and Computation Course at the Computer Science Department of the University Melbourne
- We made this material available on the Web only to ensure timely dissemination among our students, and is meant only for students' personal use.
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- V. Mascardi thanks Professor Schachte for the courtesy of sharing his teaching material on Prolog.

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1 Logical Foundations

Clause

A *clause* is a disjunction of atoms or their negations, where the variables are implicitly universally quantified.

$$A_1 \vee A_2 \vee \ldots \vee A_n \vee \neg B_1 \vee \neg B_2 \vee \ldots \vee \neg B_m$$

It is equivalent to:

$$(A_1 \vee A_2 \vee \ldots \vee A_n) \leftarrow (B_1 \wedge B_2 \wedge \ldots \wedge B_m)$$

usually written as: $(\mathit{conclusion} \leftarrow \mathit{precondition})$

$$(A_1, A_2, \ldots, A_n) \leftarrow (B_1, B_2, \ldots, B_m)$$

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Horn Clause

An Horn clause is a clause with at most one positive literal (literal = either a symbol of predicate applied to its arguments, or its negation).

Definite clause:

- rules: $A \leftarrow B_1, \ldots, B_m.$
- facts: $A.$

Negative clause:

- goal: $\leftarrow B_1, \ldots, B_m.$

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Horn Clause

Definite clauses represent "knowledge"

Logic program = set of definite clauses

Negative clauses:

- questions: $\exists X_1 \ldots X_n.(B_1 \wedge \ldots \wedge B_m)$
- negation to apply refutation

$$\neg\exists X_1 \ldots X_n.(B_1 \wedge \ldots \wedge B_m)$$

$$\forall X_1 \ldots X_n.(\neg B_1 \vee \ldots \vee \neg B_m)$$

$$\leftarrow B_1, \ldots, B_m$$

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SLD Resolution

SLD resolution

Linear resolution with a Selection function for Definite clauses.

The starting point is a given definite program P and a negative Horn clause (goal) G_0 of the form:

$$\leftarrow A_1, \ldots, A_m (m \geq 0)$$

From this goal a subgoal A_i is selected (if possible) by the selection rule. A new goal clause G_1 is constructed by selecting (if possible) some renamed program clause $B_0 \leftarrow B_1, \ldots, B_n (n \geq 0)$ whose head unifies with A_i (resulting in an mgu θ_1). If so, G_1 will be of the form:

$$\leftarrow (A_1, \ldots, Ai - 1, B1, \ldots, Bn, Ai + 1, \ldots, Am)\theta_1$$

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SLD tree

Let P be a definite program, G_0 a goal and $\mathcal{R}$ a selection rule.
The SLD tree of G_0 (using P and $\mathcal{R}$) is a (possibly infinite) labeled tree satisfying the following conditions:

- the root of the tree is labeled by G_0;
- Let $\leftarrow A_1, \ldots, A_n$ be a node and A_m the atom selected by $\mathcal{R}$. Then this node has a descendant for each clause $A \leftarrow B_1, \ldots, B_q$ of P such that A and A_m are unifiable. The descendant is:

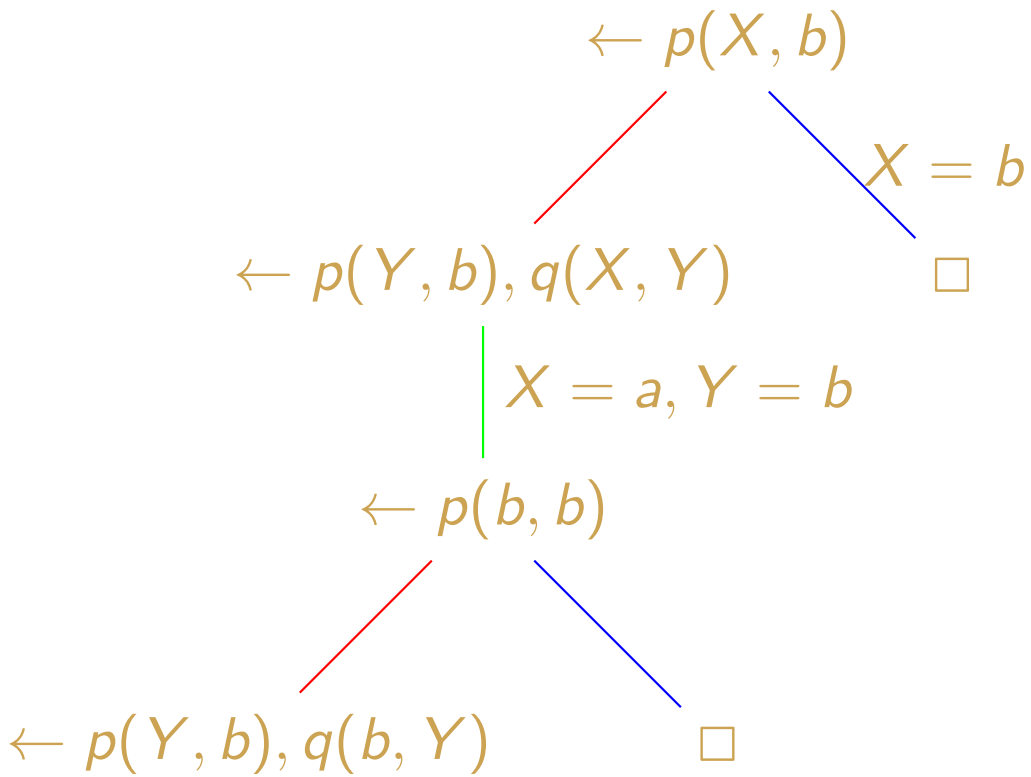
$$(A_1, \ldots, A_{m-1}, B_1, \ldots, A_{m+1}, \ldots A_n)\theta,$$

where θ is the mgu of A_m and A.

SLD tree - Example

Note that, in this example, the selection function selects the **rightmost atom**!!!

- $p(X, Z) \leftarrow p(Y, Z), q(X, Y)$
- $p(X, X)$.
- $q(a, b)$.



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